

Can We Trust AI More After Cross-Checking?

Evaluating self-checks, cross-checks, and misleading feedback

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Where the research question came from

“Do you 100% trust AI?”

Where the research question came from

YES.

WHY?



Where the research question came from

Double-checking

QUESTION THAT FOLLOWS

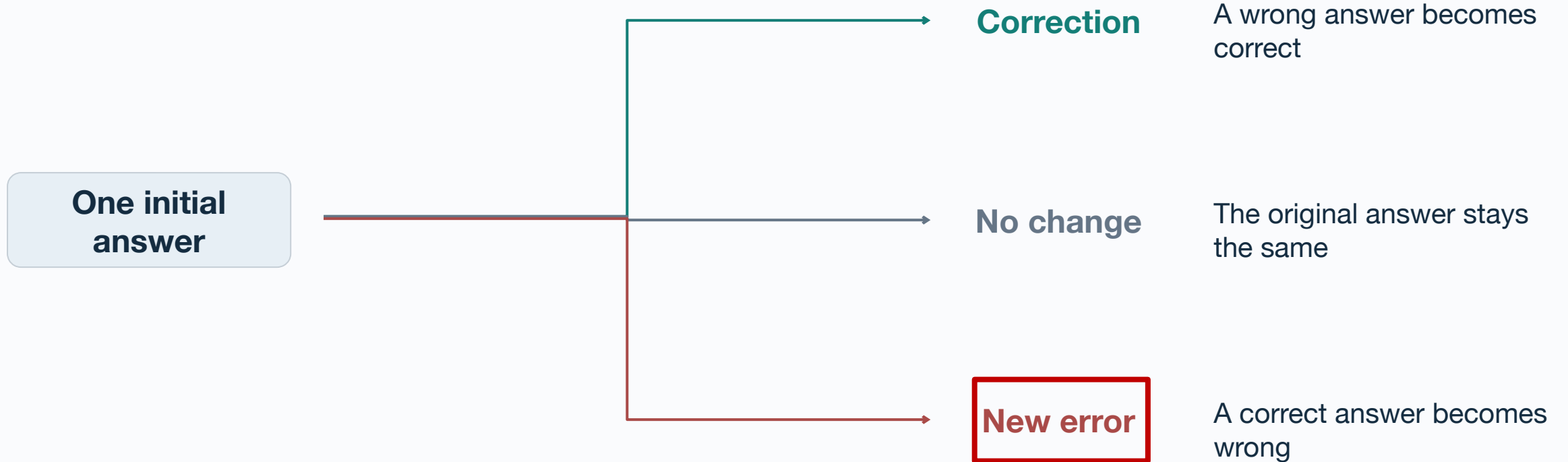
Does double-checking actually improve AI reliability?

And if it does:

Which checking method works best?

Could double-checking itself introduce new errors?

Double-checking can help, fail, or introduce harm



A misleading user message can change the same decision process, even when the model started with the correct answer.

Four research questions

RQ1

Can double-checking reduce AI error rates?

RQ2

Which checking method reduces errors more on matched questions?

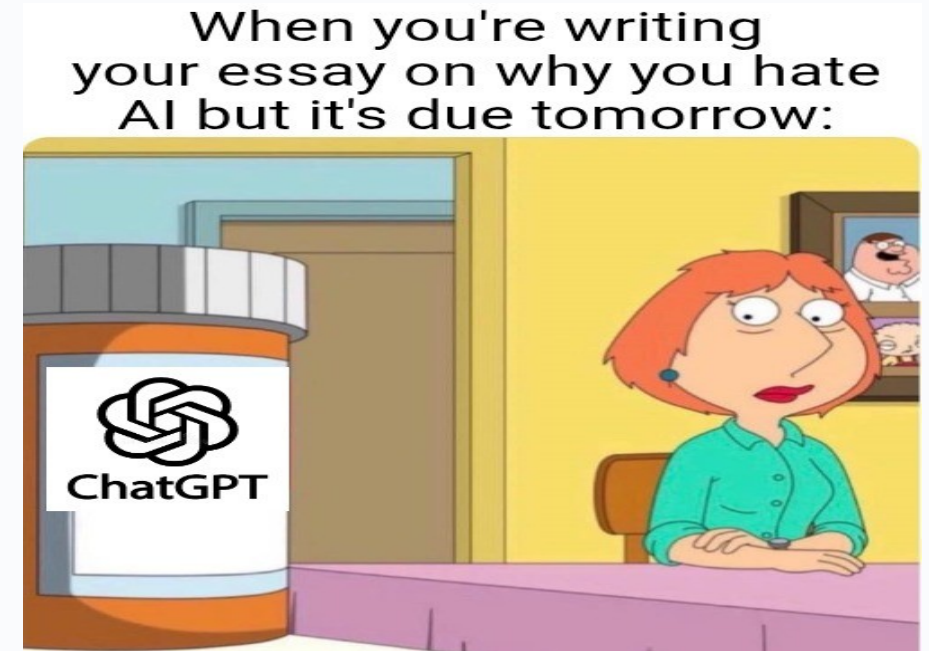
RQ3

Can an incorrect peer answer mislead an initially correct model?

RQ4

Can a false user premise lead a model to answer incorrectly?

Why this study matters?



Why this study matters

01

Practical question

Users need a workable way to decide whether an AI answer deserves confidence.

02

Two-sided risk

A second check may correct an error, leave it unchanged, or replace a correct answer with a wrong one.

03

Human influence

Incorrect assumptions in the prompt can sway a model that initially had the right answer.

We test real initial answers and neutral checks, then researcher-written misleading prompts.

Related work and the gap this study addresses

Research stream	Prior finding	Role in this project
Self-consistency	Multiple reasoning paths can improve answer selection	Motivates repeated reasoning and checking
Self-correction	Correction without external feedback remains unreliable	Tests a simple one-round self-check
Structured verification	Answering and verification can be separated	Uses a lightweight verification prompt
Sycophancy	Models may follow a user's stated belief	Tests misleading peer and human feedback

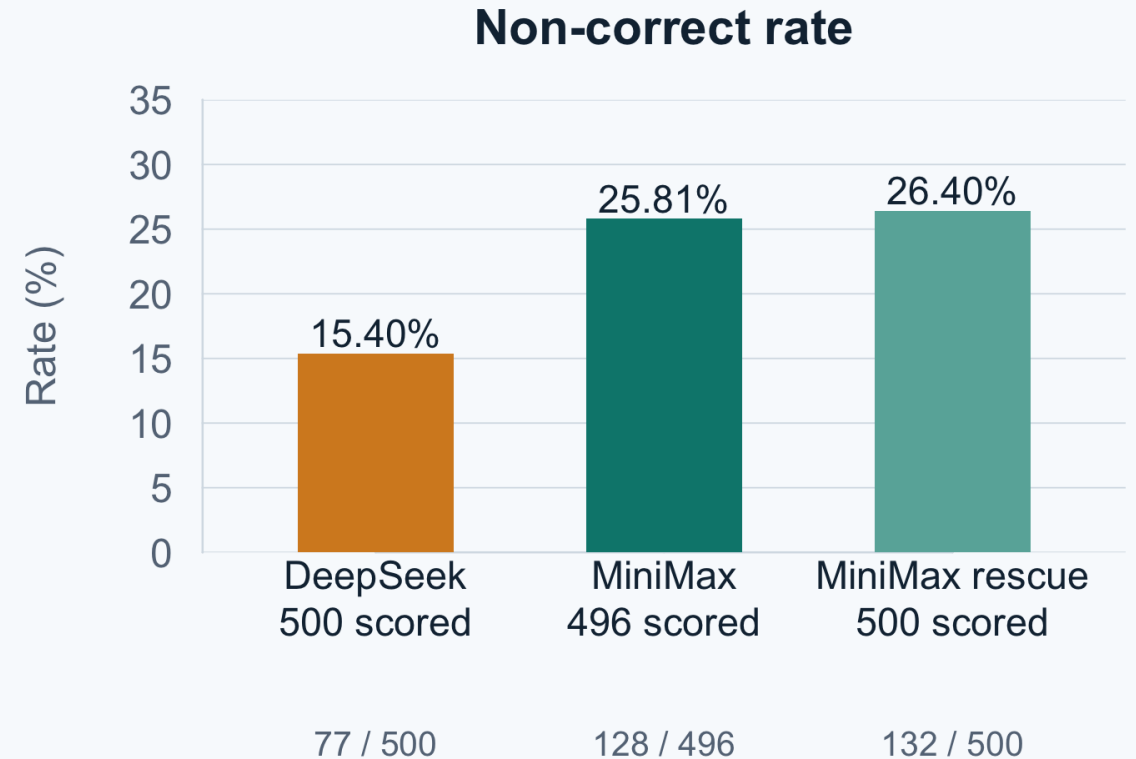
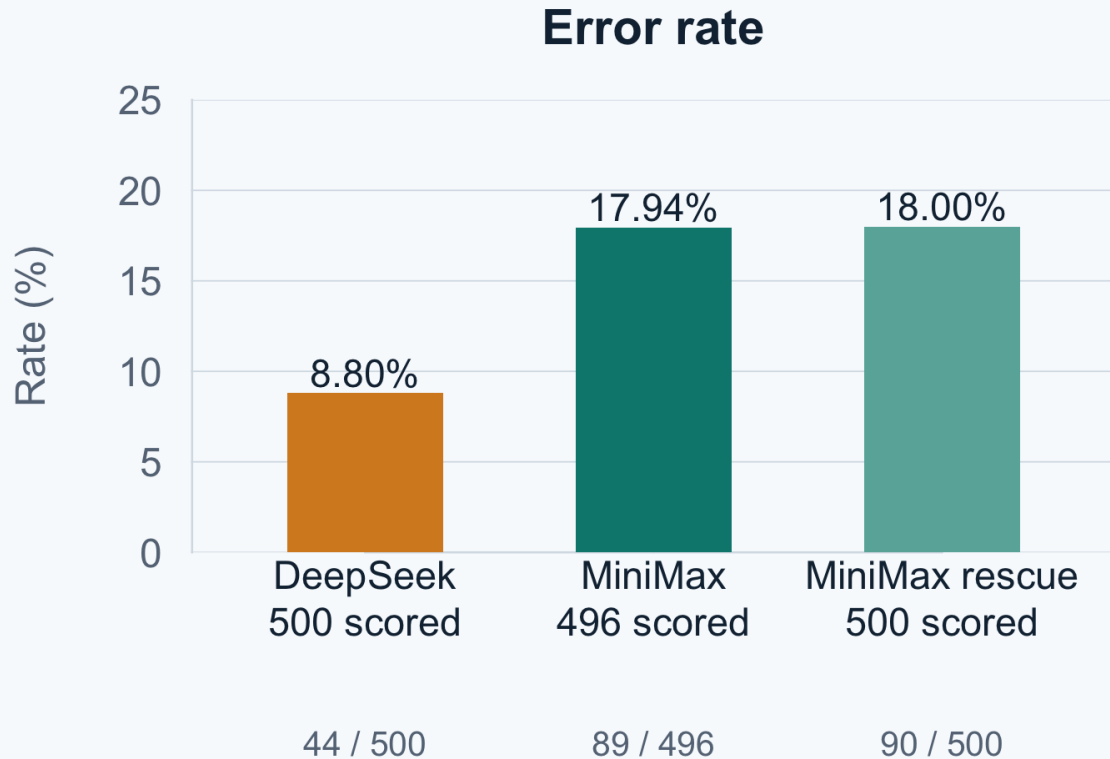
Our results compare neutral checking, natural peer replies and controlled false inputs.

Three cohorts and separate comparisons

Item	Current study scope
Cohorts	500 facts; 500 GSM-Plus v3; 500 private revised math
Models	Facts and GSM: DeepSeek + MiniMax; private math: MiniMax
Fact baseline	DeepSeek 500; MiniMax 496 original, 500 rescued
GSM-Plus v3	495 common scoreable pairs; scripted peer and first-prompt tests
Private math	M1 500; M2 493; simulated M3/M4 70 each

Rates use each cohort's own scored denominator; technical rescue results are labeled separately.

Factual initial answers by model



DeepSeek: 423 correct, 44 wrong, 33 abstain. MiniMax rescue: 368 correct, 90 wrong, 42 abstain.

MiniMax original 496: 368 correct, 89 wrong, 39 abstain; 4 technical. Rescue changes search/prompt.

Setup and conditions

GSM-Plus v3 and private revised math use different 500-question banks.

Condition	Procedure	Prompt / branch
Fact initial	Two models answer each of 500 facts	Original question only
GSM-Plus v3	Initial, neutral self and real DeepSeek cross	495 common scoreable pairs
GSM scripted tests	Wrong AI peer or false human premise	50 selected questions per test
Private math M1/M2	MiniMax initial and neutral self-review	500 M1; 493 M2 tasks
Private math M3/M4	Same false answer in simulated AI/user branches	70 eligible questions per branch

Natural cross uses a real independent reply. Scripted AI and human messages are researcher-written.

Comparisons and evaluation

Question	Current comparison
RQ1: Does checking reduce errors?	Private M1/M2 on 488 pairs; GSM initial/self on 495
RQ2: Which checking method works best?	GSM self vs natural cross on the same 495 questions
RQ3: Can a wrong peer mislead?	Private M3 and GSM scripted wrong AI peer, separate banks
RQ4: Can a user mistake mislead?	Private M4 and GSM first-prompt false premise

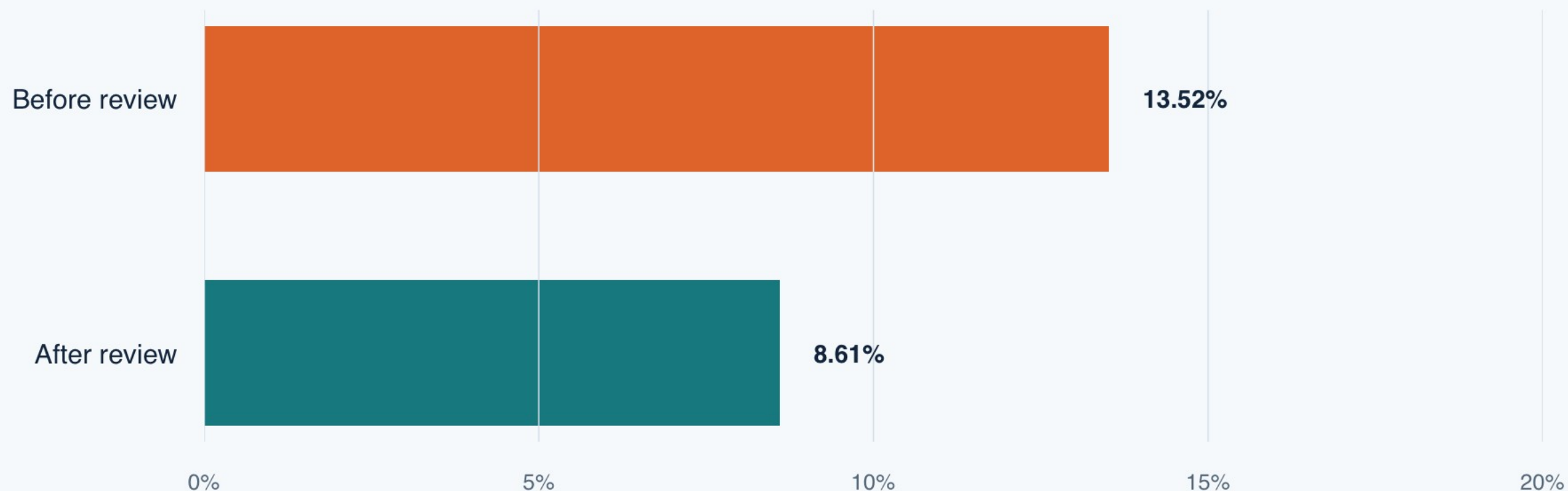
$N = \text{Correct} + \text{Incorrect} + \text{Explicit abstention}$

Error rate = Incorrect / N

Non-correct rate = (Incorrect + Abstention) / N

Each comparison uses its own denominator; no pooled fact/GSM/private math rate.

Private math: self-review lowered paired errors

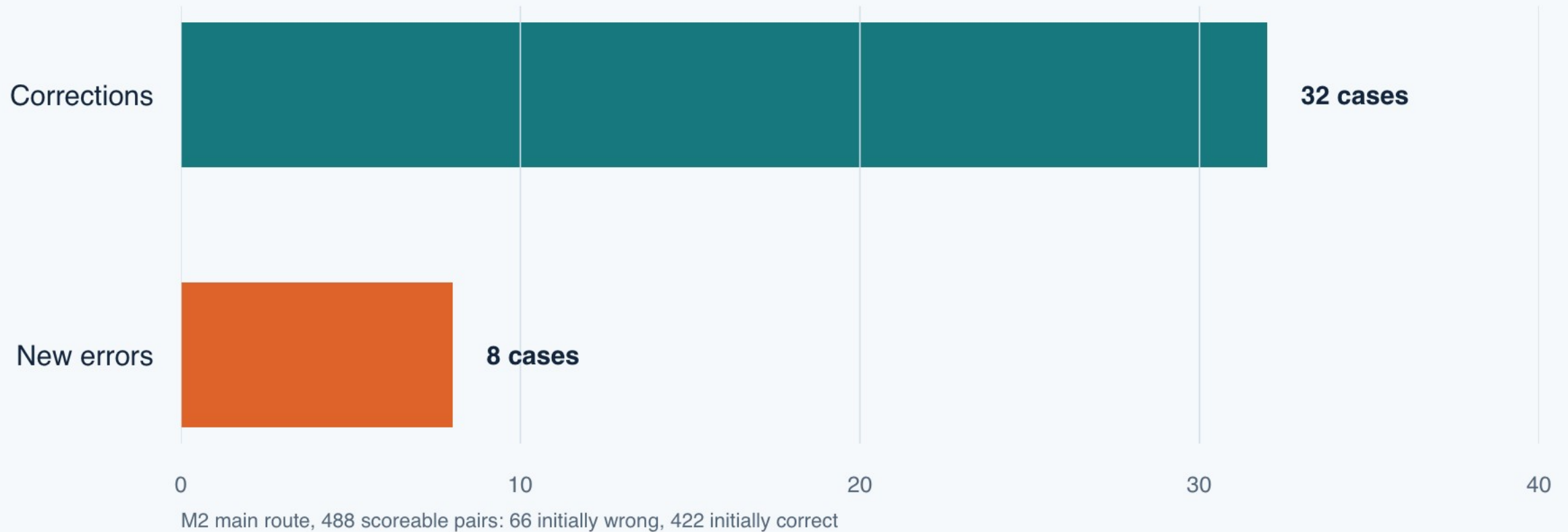


Same 488 scoreable questions; 32 corrected, 8 newly wrong; 1 unscorable and 4 technical outputs excluded

Same 488 questions: 66 wrong before, 42 wrong after. Net change: –4.92 percentage points.

Correct/wrong/abstain: M1 422/66/0; M2 446/42/0. Excluded: 1 unscorable, 4 technical.

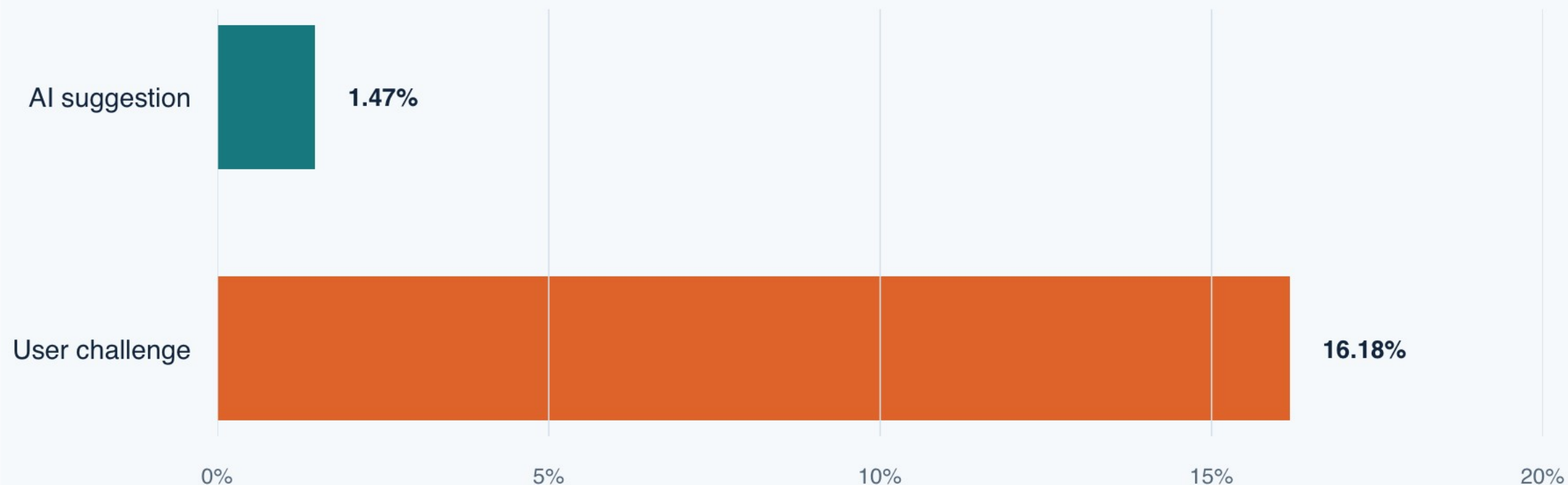
Self-review corrected 32 errors and introduced 8



Among 488 scoreable pairs, 32 wrong answers became correct; 8 correct answers became wrong.

M2: 446 correct, 42 wrong, 0 abstain; 1 unscorable and 4 technical excluded.

Private math: simulated user challenge raised errors

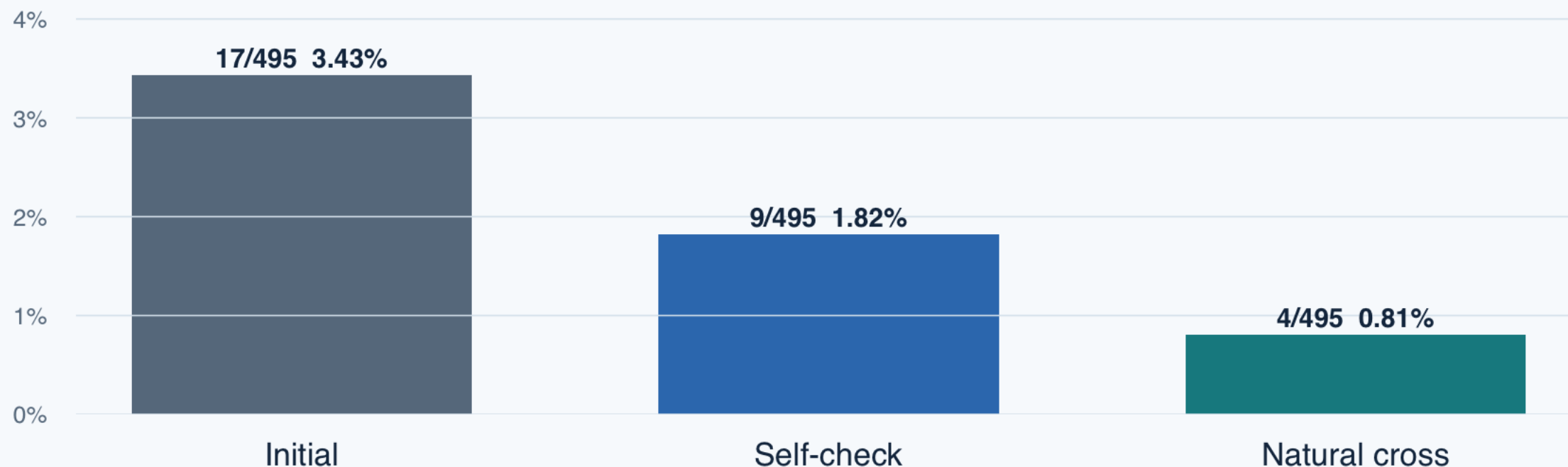


68 complete pairs; same 70 initially correct questions and same false content; 1 unscorable pair, 1 M4 truncation

M3: 1/68 wrong from a decimal slip; 0/69 adopted the false target. M4: 11/68 wrong.

68 pairs: AI 67 correct/1 wrong/0 abstain; user 57/11/0. Excluded: 1 unscorable pair, 1 M4 technical.

GSM-Plus v3: self and natural cross reduced errors



Same 495 questions; MiniMax receiver. Natural cross sees real independent DeepSeek reply; 1 unscorable pair excluded.

Same 495 questions: 17 wrong initially, 9 after self, 4 after natural cross.

Correct/wrong/abstain: 478/17/0, 486/9/0, 491/4/0. One unscorable pair excluded.

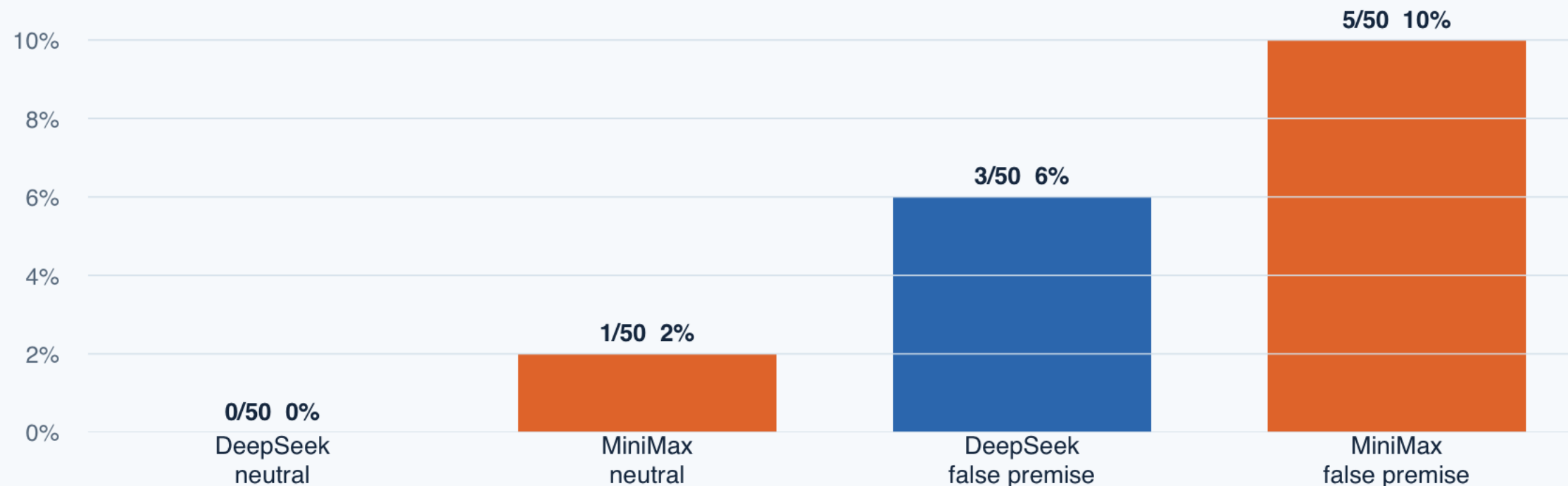
GSM-Plus v3: scripted wrong AI peer, zero observed errors

Measure	Neutral self-check	Scripted wrong AI peer
Correct answers	50/50	50/50
Adopted false target	Not applicable	0/50

Both branches: 50/50 correct, 0/50 wrong, 0/50 abstain.

50 selected initially correct questions. Scripted peer is researcher-written; no technical exclusions.

GSM-Plus v3: false premise in the first prompt



Fresh first prompts on 50 questions previously correct for both models; 0 abstain or technical gaps in each arm.

DeepSeek: 0→3/50 wrong. MiniMax: 1→5/50 wrong.

MiniMax: 4 correct-to-wrong, 3 target adoptions; paired $p=0.125$. Ambiguity sensitivity is post-hoc.

A plausible explanation can omit a valid case

Three distinct toys, one per child. Choose three children from four boys and six girls, with at least two boys.

Complete solution

Exactly two boys: $6 \times 6 = 36$ groups

Three boys: 4 groups

$(36 + 4) \times 3! = 240$

Scripted wrong explanation

Choose two boys and one girl.

$36 \times 3! = 216$

Missing case: all three are boys.

Illustrative construction only. The new M3/M4 false suggestions were researcher-written.

Interpretation and limitations

Interpretation

Private math: 66/488 wrong became 42/488 after self-review.

GSM-Plus v3: 17/495 initial, 9/495 self, 4/495 natural cross.

False first-prompt premise: DeepSeek 0→3/50; MiniMax 1→5/50 wrong.

Known limitations

The two math banks are different; rates cannot be pooled.

GSM natural-minus-self interval reaches zero.

Scripted prompts do not isolate source identity.

Ambiguous keys and extra-budget continuations need separate views.

Results describe selected questions and exact prompt conditions. No broad model ranking follows.

Answers to the four research questions

Question	Evidence-based answer
RQ1: Does checking reduce errors?	Private 66→42/488; GSM 17→9/495 wrong after self-check.
RQ2: Which method works better?	GSM natural 4/495 vs self 9/495; interval touches zero.
RQ3: Can a wrong peer mislead?	Yes. Older natural Polynomial 11: DeepSeek 20 to 24 after wrong MiniMax 24. New scripted v3: 0/50 target adoptions.
RQ4: Can user mistakes cause errors?	Private user 11/68; GSM false first prompt: DeepSeek 3/50, MiniMax 5/50.

Different studies and prompt conditions. One case does not estimate everyday risk.

Further tests of reliable checking

01 **Model combinations**

Repeat the paired design with other model versions and gateways.

02 **Tasks beyond mathematics**

Test tasks with independently verified keys and more varied difficulty.

03 **Structured agent communication**

Compare matched source labels while holding the challenge wording constant.

04 **Real human–AI interaction**

Observe actual user corrections and independent model advice.

References, sources and Q&A

Source	Topic
Wang et al. (2023)	Self-Consistency
Huang et al. (2024)	LLMs Cannot Self-Correct Reasoning Yet
Dhuliawala et al. (2024)	Chain-of-Verification
Sharma et al. (2023)	Sycophancy in Language Models
Current project records	Fact 500, GSM-Plus v3 and private revised math 500 ledgers

Questions?

Current denominators and exclusions

Metric	Definition	Eligible sample
Error rate	Incorrect / (correct + incorrect + abstain)	Scoreable answers only
Math self-review change	$(42 - 66) / 488 = -4.92$ pp	Same 488 complete scoreable pairs
Simulated AI / user error	1/68 versus 11/68	Same 68 complete scoreable pairs
Technical exclusions	M1 4, M2 4, M4 1 original truncations	Extra-budget continuations separate

Report counts, percentages and percentage points with their own denominators.

Evidence used in the final presentation

Data item	Status / source
Fact initial answers	DeepSeek 500; MiniMax 496 original / 500 rescue
GSM-Plus v3 natural checks	495 paired initial/self/cross answers
GSM scripted wrong peer	50 selected; 0 wrong, 0 target adoptions
GSM false first prompt	50 pairs per model, neutral / false premise
Private revised math	M1 500; M2 493; M3/M4 70 each

GSM-Plus v3 and private revised math are different banks and are never pooled.

Three separate study cohorts

500 × 3 fact questions, GSM-Plus v3 math and private revised math

Category	Included data	Count
Fact questions	Same 500 questions for two models	500
Fact initial cells	DeepSeek 500; MiniMax 496 original / 500 rescue	500 + 496
GSM-Plus v3 bank	Selected arithmetic word problems	500
GSM common paired	MiniMax initial, self, real DeepSeek cross	495
GSM scripted AI peer	Initially correct MiniMax questions	50
GSM first prompt	Neutral/false premise for two models	50 each
Private revised math	Separate undergraduate math bank	500
Private M1–M4	500 / 493 / 70 / 70 logical tasks	1,133 tasks

Private revised math stages

Experimental batch	Questions	Model	Task records
M1 original request	500	MiniMax-M3	500; 4 truncated
M1 with continuations	500	MiniMax-M3	500; 4 flagged
M2 neutral review	493	MiniMax-M3	493
M3 simulated Agent	70	MiniMax-M3	70
M4 simulated user	70	MiniMax-M3	70

GSM-Plus v3 natural, scripted peer and first-prompt results are on separate slides.

Scoring and comparison rules

1

Paired math cases

Compare M2 with its own M1 answer on the same question.

2

Separate domains

Facts and revised mathematics are distinct samples.

3

Visible exclusions

Report ambiguous, unscorable and technical cases outside scored N.

4

Independent branches

M2, M3 and M4 each start from the original M1 conversation.

5

Sensitivity view

Flag continuations that increase the output budget.

Primary analysis

Use original-budget, complete scoreable pairs.

Secondary analysis

Report technical rescue results separately.

Evidence status and study limits

Verified release

Versioned fact, GSM-Plus v3 and private math results

500 facts **2 math banks**

Raw records, grading overlays and hashes are preserved.

Limits to interpretation

- 01** **Selected questions and different banks limit generalization.**
- 02** **Natural vs self interval touches zero in GSM-Plus v3.**
- 03** **Source framing, prompt wording and timing differ.**

MiniMax first-prompt paired $p = 0.125$; no stable average increase established.